

Sanchit Chhabra

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SUMMARY

Graduate student specializing in Artificial Intelligence with experience building AI-powered applications using Python, FastAPI, LangChain, and vector databases. Skilled in machine learning, Retrieval-Augmented Generation (RAG), and backend development, with a passion for developing scalable AI solutions that transform data into intelligent user experiences. Seeking Co-op opportunities.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript
Artificial Intelligence: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LangChain, Prompt Engineering
Machine Learning: Scikit-learn, Random Forest, K-Means Clustering, Hierarchical Clustering, PCA, Supervised & Unsupervised Learning
Data Science: Pandas, NumPy, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation
Backend Development: FastAPI, REST APIs, Spring Boot, API Integration
Databases & Vector Stores: PostgreSQL, MySQL, ChromaDB, Vector Embeddings
Cloud & Deployment: Render, Vercel, Git, GitHub
Tools: Jupyter Notebook, VS Code, Swagger, JIRA, Postman

PROJECTS

Personalized Learning and Academic Support Platform, *FastAPI + Angular*

- Developed an AI-powered academic support platform using Retrieval-Augmented Generation (RAG)
- Integrated Large Language Models (LLMs) with LangChain for intelligent question answering
- Implemented vector embeddings and semantic search using ChromaDB for efficient knowledge retrieval
- Developed RESTful APIs using FastAPI to support AI-powered interactions
- Processed, chunked, and indexed educational content using vector embeddings to improve retrieval accuracy and reduce hallucinations
- Deployed backend services using Render and frontend using Vercel
- Collaborated within an Agile team using Git, GitHub, and JIRA

Unsupervised Machine Learning & Cluster Analysis, *Python + Scikit-learn + PCA*

- Performed exploratory data analysis (EDA) and data preprocessing on the Iris dataset
- Applied feature scaling and dimensionality reduction using Principal Component Analysis (PCA)
- Implemented K-Means and Hierarchical Clustering to identify natural data groupings
- Evaluated clustering performance using Silhouette Score and Adjusted Rand Index (ARI)
- Visualized clustering results using scatter plots, dendrograms, and confusion matrices
- Compared clustering algorithms to analyse model performance and cluster quality

Housing Affordability Analysis, *Tableau + SQL + Data Storytelling*

- Analyzed Canadian housing affordability using datasets from CMHC and Statistics Canada
- Cleaned and transformed multi-source datasets to improve data quality and consistency
- Built interactive Tableau dashboards using bubble, line, and bullet charts to visualize rent, income, and affordability trends
- Identified affordability trends by comparing rental costs and household income across Canadian regions
- Presented findings through data storytelling techniques to support data-driven decision-making

WORK EXPERIENCE

Wristkey Global Solutions, *Data & Risk Analyst* USA

- Analysed security and operational risks across a mobile and wearable wallet ecosystem
- Collected and evaluated data related to NFC communication and biometric authentication
- Produced analytical reports summarising vulnerabilities, trends, and mitigation strategies
- Conducted competitive market analysis by benchmarking security features against industry solutions
- Presented findings and recommendations to support strategic business decisions

Bumima Trustshare, *Risk & Compliance Analyst* USA

- Analyzed security risks and categorised incidents based on business impact and severity
- Collected and documented operational data to support SOC 2 compliance initiatives
- Developed standard operating procedures (SOPs) and process documentation for incident management
- Collaborated with stakeholders to improve reporting processes and operational efficiency
- Produced analytical reports supporting risk assessment and business decision making

CERTIFICATES

• CompTIA CysA+ (CS0-003) 

EDUCATION

Artificial Intelligence Analysis, Design and Implementation, *Durham College* 01/2026 – Present | Oshawa, Canada
Related Coursework: AI Algorithms, Introduction to AI, Studies in AI, Data Visualization and Storytelling

Cybersecurity Operation, *York University* 01/2025 – 12/2025 | Toronto, Canada

Bachelor of Science (Computer Science), *ARSD Delhi University* 11/2021 – 06/2024 | Delhi, India